

Dashboard 3: Borrower Risk & Loan Delinquency Analysis

Introduction:

Analyzing **borrower risk profiles** and **loan delinquency trends** is essential for effective **credit risk management**. By segmenting borrowers based on **risk levels**, **repayment behavior**, and **overdue amounts**, the analysis provides valuable insights into **loan performance** and **default patterns**. These insights support the identification of **high-risk loans**, the enhancement of **collection strategies**, and the optimization of **lending decisions** to maintain a **healthy loan portfolio**.

DAX Queries:

1. Total Delinquent Loan:

```
TotalDelinquentLoans =  
CALCULATE (  
    COUNT (bondora_data[loanid]),  
    bondora_data[principaloverduebyschedule] > 30  
)
```

2. Delinquency Rate (%):

```
Delinquency Rate =  
DIVIDE (  
    [TotalDelinquentLoans],  
    COUNT (bondora_data[loanid]),  
    0  
)
```

3. Average Days Past Due:

```
AvgDaysPastDue = AVERAGE  
(bondora_data[principaloverduebyschedule])
```

4. Average Days to Recover:

```
AverageDays_to_Recover =  
AVERAGEX (  
    FILTER (  
        bondora_data,  
        NOT ISBLANK (bondora_data[lastpaymenton] )  
        && NOT ISBLANK  
(bondora_data[stageactivesince] )  
    ),  
    ABS (  
        DATEDIFF (  
            bondora_data[stageactivesince],  
            bondora_data[lastpaymenton],  
            DAY  
        )  
    )  
)
```

5. High-Risk Borrower Percentage (%):

```
High_Risk_Borrowers(%) =  
DIVIDE (  
    COUNTROWS (  
        FILTER (  
            'bondora_data',  
            'bondora_data'[creditscoreeemini] < 600  
        )  
    ),  
    COUNT('bondora_data'[loanid]),  
    0  
)
```

Visualizations:

Visualization	Visualization Type
Total Loan Amount Issued	KPI Card
Total Delinquent Loans	KPI Card
Average Days past Due	KPI Card
Delinquency Rate (%)	KPI Card
High Risk Borrowers (%)	KPI Card
Delinquency Rate by Loan Category	Bar Chart - X-axis: Delinquency_Rate_byCategory - Y-axis: rating
Distribution of Loans by Risk Category	Pie Chart Rating and count of loanid
Delinquency Summary by Loan Category	Table: Columns: Loan Category, Total Loans, Delinquent Loans, Delinquency Rate (%)
Delinquency Rate Trend over Time:	Line Chart: - X-axis: Year - Y-axis: Delinquency Rate
Distribution of Days past Due:	Histogram: - X-axis: principaloverduebyschedule(bins) - Y-axis: Count(loanid)
High_Borrowing_risk_Loanpurpose:	Table - X-axis: useofloan - Y-axis: High_Risk_Borrowers(%)

Filters & Slicers:

- useofloan
- Delinquency Category
- creditscoreemini
- loanduration(months)
- amount
- Country

Conclusion:

Analyzing **borrower risk** and **loan delinquency trends** provides valuable insights into overall credit performance. By tracking key **KPIs** such as **delinquency rates**, **overdue amounts**, and **borrower credit risk**, the analysis enhances visibility into lending operations. Interactive **visualizations** enable borrower segmentation and reveal **overdue patterns**, improving the precision of **risk assessment**. Furthermore, the use of **filters and slicers** facilitates detailed exploration across **credit scores**, **loan categories**, and **regions**, supporting more informed and strategic **lending decisions**.